RELIANCE GRANITO

Export Documentation Management System — Developer Workflow

1. Main shipment-to-document workflow

1. Login & Dashboard	→	2. Create Shipment	→	3. Add Shared Data	→	4. Create Invoice	→	5. Shipping Bill	→	6. Create BL
User signs in; role and permissions are checked.		Generate unique shipment ID; select exporter and buyer.		Parties, goods, containers, weights, ports and route.		Enter items, HSN, quantity, rates, currency and terms.		Enter/import customs details and attach official source PDF.		Enter carrier, vessel, ports, container, packages and weights.

Shared-data principle: one shipment links the commercial invoice, shipping bill and bill of lading (BL). Reuse shared data where appropriate, but keep document-specific fields independent.

2. Validation and approval workflow

7. Cross-document Validation	→	8. Discrepancy Review	→	9. Review & Finalize
Compare invoice ↔ shipping bill ↔ BL: item quantities, values, HSN, invoice/SB references, containers, packages and weights.		Show field-level mismatches. User reviews and records a resolution; do not silently overwrite source values.		Reviewer checks required fields and approves a new immutable document version.

3. Document output workflow

Original	or	Edited	or	HIDE	→	Preview & Export
Preserve the original uploaded file/data version.		Create an edited version; retain previous values and audit history.		Hide only fields mapped from highlighted regions in the reference PDF. Keep underlying data.		Render PDF, inspect layout, download/print; save version, template ID and checksum.

4. Key system rules

- **Official customs document:** retain the actual customs-issued shipping bill unchanged. Application-generated previews must not be presented as official customs-issued documents.
- **Version control:** finalized versions are immutable. Corrections create a new version; never overwrite the original source PDF.
- **HIDE:** use a per-template field visibility map based on the highlighted regions in the supplied HIDE PDFs; do not delete data.
- **Amounts and quantities:** flag inconsistencies for review. Do not automatically recalculate or overwrite fixed totals without confirmation and audit logging.
- **Security:** enforce role permissions server-side; store PDFs privately; log edits, approvals, downloads where required, and template changes.

5. Data model and relationships

Core relationship: Shipment → Commercial Invoice(s) → Invoice Items; Shipment → Shipping Bill(s) → Shipping Bill Items; Shipment → BL(s) → BL Containers. A shipment may also contain multiple shipment containers, parties, attachments, declaration records, events and generated document versions.

Table	Main purpose / key fields
shipments	shipment_number, exporter_party_id, buyer_party_id, status, created_by
parties	party_type, name, address, IEC, GSTIN, tax_code, contact details
commercial_invoices	invoice_number, invoice_date, currency, invoice_total, weights, status
commercial_invoice_items	invoice_id, description, HSN, quantity, UQC, rate, line_amount
shipping_bills	shipment_id, SB number/date, port code, valuation, source_pdf_path, status
shipping_bill_items	SB item fields: HSN, quantity, rate, FOB INR, PMV, drawback, IGST, scheme
bill_of_ladings	BL number, carrier, vessel/voyage, ports, dates, freight terms
shipment_containers / bl_containers	container number, seal, package count, gross/net weight
document_templates / document_template_fields	document type, variant, template version, page, coordinates, visibility
document_versions	source record, variant, version number, snapshot, PDF path, checksum, generated_by
audit_logs	user, entity, action, old/new values, timestamp, shipment ID
shipping_bill_events / declarations / attachments	Customs events, supporting document metadata, declarations and private file paths

6. Build and test sequence

Phase	Deliverables	Exit check
1. Foundation	Auth, roles, masters, shipments, migrations and relationships	Shipment and party CRUD; permissions enforced
2. Invoice	Invoice editor, item calculations, original/edited PDF templates	PDF matches supplied invoice; overrides audited
3. Shipping Bill	Six-page field mapping, item/declaration data, source PDF upload	All pages mapped; source file preserved
4. BL and HIDE	BL editor/template, highlighted-field visibility maps	Unmarked fields remain visible; layout checked
5. Audit & release	Version history, discrepancies, security, backup and regression tests	Acceptance checklist passed; recovery tested

Open verification items: verify the complete page-by-page mapping for all six shipping bill pages; visually confirm each highlighted HIDE field; manually verify ambiguous HSN digits and all original/edited value differences. OCR text alone is not authoritative for ambiguous characters.